

Ejercicio 1:

```
package com.utad.poo.tema2;
public class Ejemplo {
    public static void main(String[] args) {
        String shortMsg = "Short Message";
        String longMsg = "Looooooooooooooooong Message";
        boolean printShort = true;
        int j = 0;
        int counter; //Da error ya que no esta inicializada
        //System.out.printf("Valor Inicial de Counter: %d\n", counter);
        if (printShort) {
            System.out.println(shortMsg);
        }
        else {
            System.out.println(longMsg);
        }

        System.out.println("This message is always printed");

        for (int i = 0; i < 10 ; i++) {
            System.out.println("i: " + i + " Are we there yet?");
        }

        while (j < 7) {
            System.out.println("j: " + j + " I'm going to pass this course");
            j++;
        }
    }
}

Short Message
This message is always printed
i: 0 Are we there yet?
i: 1 Are we there yet?
i: 2 Are we there yet?
i: 3 Are we there yet?
i: 4 Are we there yet?
i: 5 Are we there yet?
i: 6 Are we there yet?
i: 7 Are we there yet?
i: 8 Are we there yet?
i: 9 Are we there yet?
j: 0 I'm going to pass this course
j: 1 I'm going to pass this course
j: 2 I'm going to pass this course
j: 3 I'm going to pass this course
j: 4 I'm going to pass this course
j: 5 I'm going to pass this course
j: 6 I'm going to pass this course
```

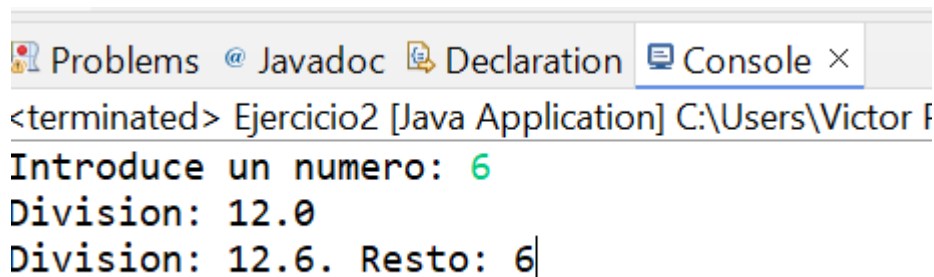
Ejercicio 2:

```
package com.utad.poo.tema2;
import java.util.Scanner;
public class Ejercicio2 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        double num = 0;
        double mult = 0;
        double div = 0;
        int resto = 0;

        System.out.print("Introduce un numero: ");
        num = input.nextDouble();
        input.close();

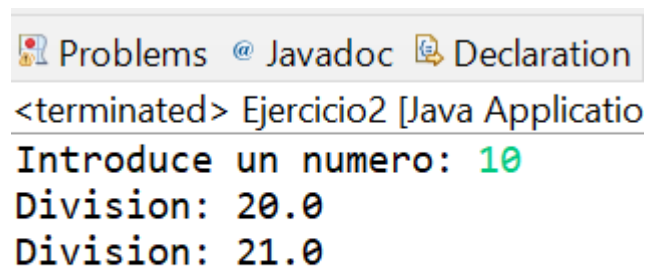
        mult = num * 20;
        div = mult / 10;
        System.out.println("Division: "+ div);

        mult += num;
        div = mult / 10;
        resto = (int)(mult % 10);
        if (resto != 0) {
            System.out.println("Division: "+ div + ". Resto: "+ resto);
        }
        else {
            System.out.println("Division: "+ div);
        }
    }
}
```



The screenshot shows the IDE's console window with tabs for Problems, Javadoc, Declaration, and Console. The output of the program for the input '6' is displayed as follows:

```
<terminated> Ejercicio2 [Java Application] C:\Users\Victor F
Introduce un numero: 6
Division: 12.0
Division: 12.6. Resto: 6|
```



The screenshot shows the IDE's console window with tabs for Problems, Javadoc, Declaration, and Console. The output of the program for the input '10' is displayed as follows:

```
<terminated> Ejercicio2 [Java Applicatio
Introduce un numero: 10
Division: 20.0
Division: 21.0
```

Ejercicio 3:

```
package com.utad.poo.tema2;
```

```
public class Ejercicio3 {
```

```
    public static final String BLACK = "B";
```

```
    public static final String WHITE = "W";
```

```
    public static final int TAM = 8;
```

```
    public static void main(String[] args) {
```

```
        for (int i = 0; i < Ejercicio3.TAM; i++) {
```

```
            for (int j = 0; j < Ejercicio3.TAM; j++) {
```

```
                if ((i+j)%2 == 0) {
```

```
                    System.out.print(Ejercicio3.BLACK);
```

```
                }
```

```
                else {
```

```
                    System.out.print(Ejercicio3.WHITE);
```

```
                }
```

```
            }
```

```
            System.out.println();
```

```
        }
```

```
    }
```

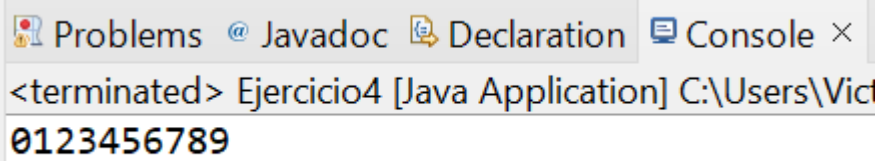
```
}
```

```
Problems @ Javadoc Declaration Console ×  
<terminated> Ejercicio3 [Java Application] C:\Users\Victor Pei  
BWBWBWBW  
WBWBWBWB  
BWBWBWBW  
WBWBWBWB  
BWBWBWBW  
WBWBWBWB  
BWBWBWBW  
WBWBWBWB
```

Ejercicio 4:

```
package com.utad.poo.tema2;
public class Ejercicio4 {
    public static void main(String[] args) {
        int i = 0;

        while (i < 10) {
            System.out.print(i);
            i++;
        }
        System.out.println();
    }
}
```



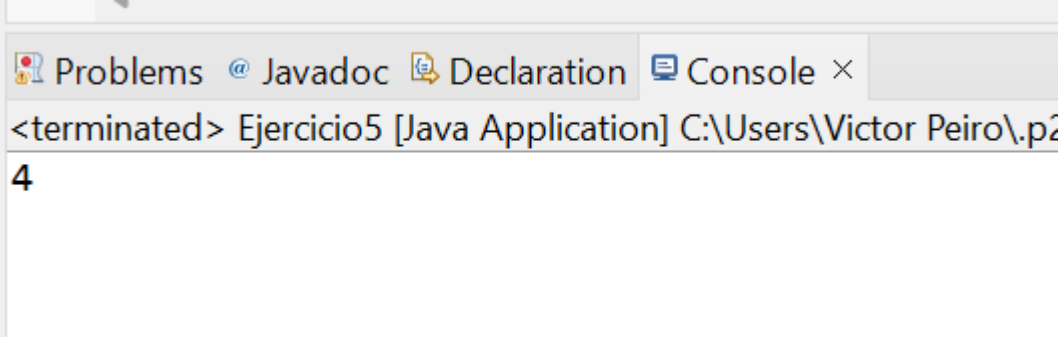
The screenshot shows an IDE window with tabs for Problems, Javadoc, Declaration, and Console. The Console tab is active, displaying the output of the program: 0123456789. The text is preceded by a terminated status message.

```
<terminated> Ejercicio4 [Java Application] C:\Users\Vict
0123456789
```

Ejercicio 5:

```
package com.utad.poo.tema2;
public class Ejercicio5 {
    public static final int MAX_ITEMS = 10;
    public static void main(String[] args) {
        int [] items = { 2, 4, 6, 8, 10, 9, 7, 5, 3, 1};
        boolean found = false;
        int find = 10;
        int i;

        for (i = 0; i < MAX_ITEMS && !found; i++) {
            found = (items[i] == find);
        }
        System.out.println(i-1);
    }
}
```



The screenshot shows an IDE console window with tabs for Problems, Javadoc, Declaration, and Console. The Console tab is active, displaying the output of the Java application. The text in the console is: <terminated> Ejercicio5 [Java Application] C:\Users\Victor Peiro\.p2 4. The number 4 is printed on a new line below the previous text.

```
<terminated> Ejercicio5 [Java Application] C:\Users\Victor Peiro\.p2
4
```

Ejercicio 6:

```
package com.utad.poo.tema2;
public class Ejercicio6 {
    public static final int NUM = 4;

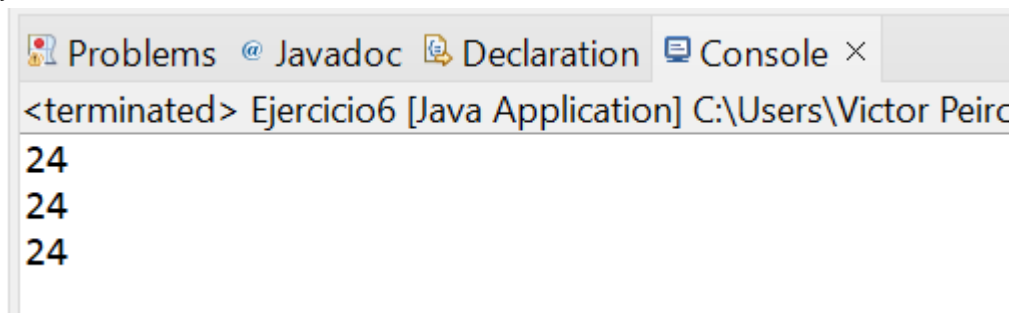
    public static int factorial(int x) {
        if (x <= 1) {
            return 1;
        }
        else {
            return x*factorial(x-1);
        }
    }

    public static void main(String[] args) {
        int result = 1;
        int j = 1;

        for (int i = 1; i <= Ejercicio6.NUM; i++) {
            result *= i;
        }
        System.out.println(result);
        result = 1;

        while (j <= Ejercicio6.NUM) {
            result *= j;
            j++;
        }
        System.out.println(result);

        result = Ejercicio6.factorial(Ejercicio6.NUM);
        System.out.println(result);
    }
}
```

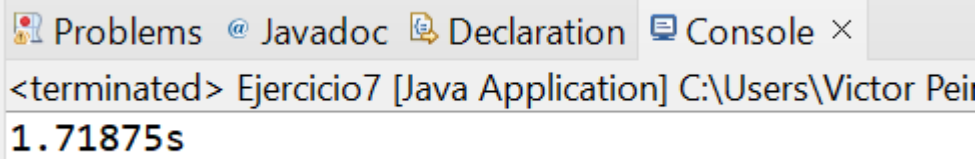


The screenshot shows an IDE window with tabs for 'Problems', 'Javadoc', 'Declaration', and 'Console'. The 'Console' tab is active, displaying the output of the program. The output consists of three lines, each showing the number '24', which is the result of calculating the factorial of 4. The first line is preceded by '<terminated>'. The console title bar indicates the application is 'Ejercicio6 [Java Application]' and the file path is 'C:\Users\Victor Peirc'.

```
<terminated> Ejercicio6 [Java Application] C:\Users\Victor Peirc
24
24
24
```

Ejercicio 7:

```
package com.utad.poo.tema2;
public class Ejercicio7 {
    public static double calcTiempo(double x0, double v, double x) {
        return (x - x0)/v;
    }
    public static void main(String[] args) {
        double tiempo = 0;
        tiempo = Ejercicio7.calcTiempo(0, 3.2, 5.5);
        System.out.println(tiempo + "s");
    }
}
```



The screenshot shows the 'Console' tab of an IDE. The text in the console is:
<terminated> Ejercicio7 [Java Application] C:\Users\Victor Pei
1.71875s